

Language Context Shape the structures of LLM-generated networks

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Introduction

- LLM-base social simulation have been applied in a wide range of research, such as political science, economics, and communication studies [1] [2]
- Some studies suggests that LLM agents are able to generate networks structurely similar to real-world ones [3].
 - However, other points out biases, especially when research asking LLM agents play minority roles. LLM's value might also closer to specific cultural group [4][5].
- Due to inqual distribution of language in training datasets, LLMs generally perform better in english [6][7]
 - Most of research on multilingual prompt focus on how to improve LLM's performance on minority languages. Rarely does research examine the relationship between prompt language and LLM's behaviour in social simulation.
- In this research, I examine whether prompt language LLM agent's behavior, leading to different network structures
 - Does prompt language affect LLM's behaviour in social simulation?
 - Does the variance in LLM's behaviour across different country contexts accurately reflect the differences between countries in real world?

Methods: Real-world data

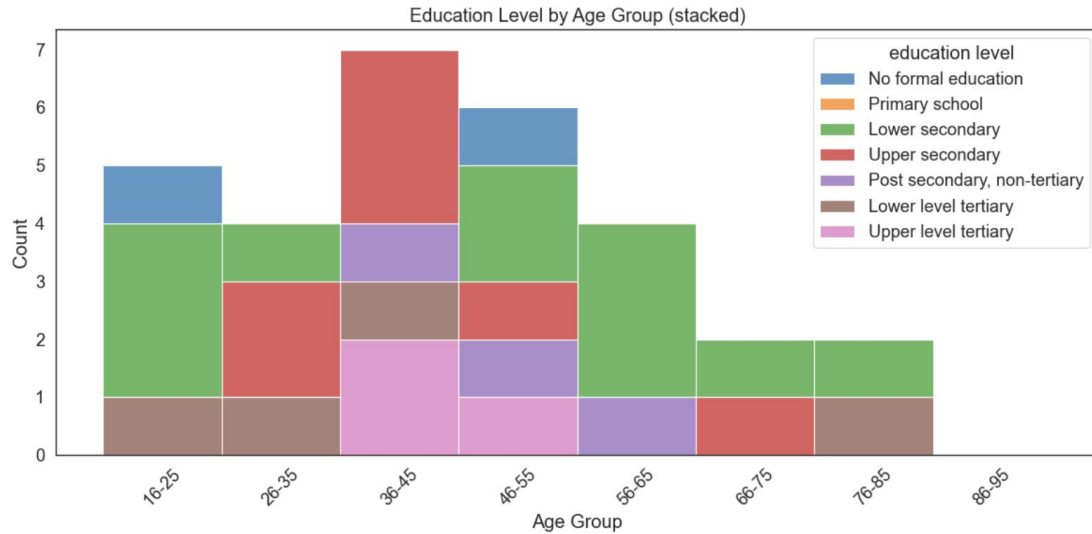
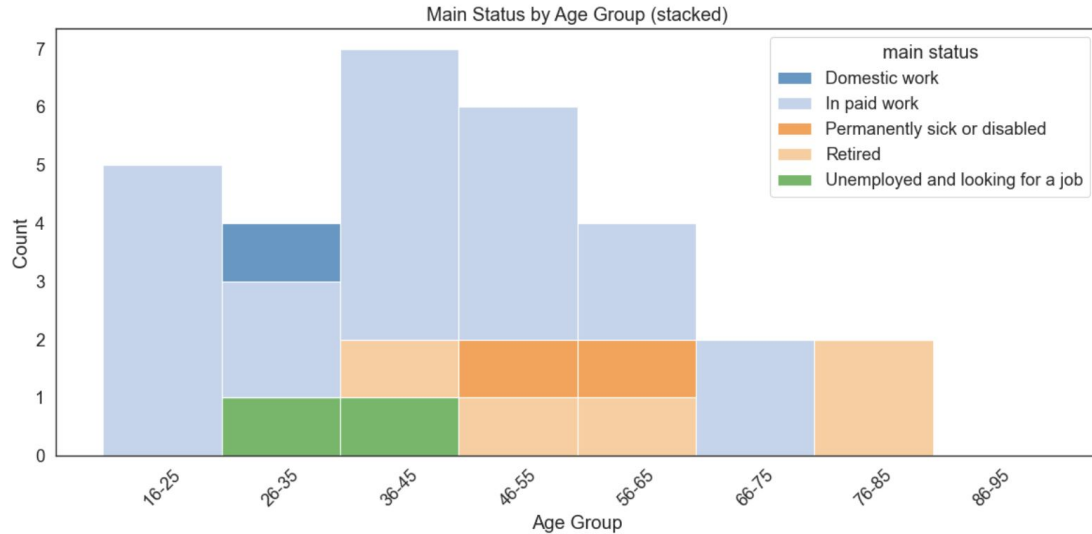
- ISSP 2017 - Social Networks and Social Resources
 - [The International Social Survey Programme](#)
 - This is a **cross-national** survey that collects questionnaire data from individuals in different countries.
 - It Includes basic demographic variables (e.g. gender and age) and other aspects of social life (e.g. level of social trust, loneliness ... etc)
 - In total 30 countries

2.3 Geographic coverage

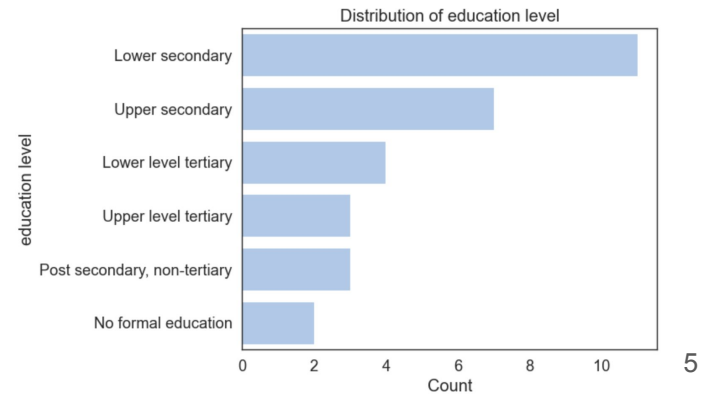
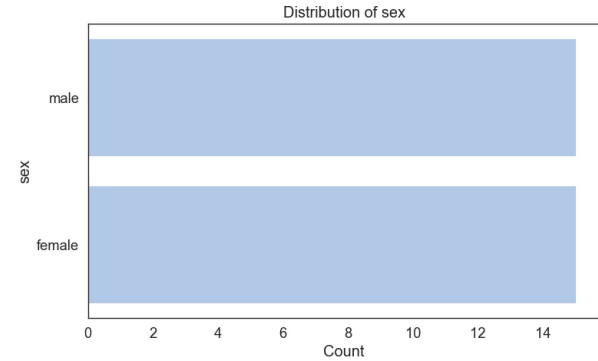
Austria (AT); Australia (AU); Switzerland (CH); China (CN); Czech Republic (CZ); Germany (DE); Denmark (DK); Spain (ES); Finland (FI); France (FR); Great Britain (GB-GBN); Croatia (HR); Hungary (HU); Israel (IL); India (IN); Iceland (IS); Japan (JP); Lithuania (LT); Mexico (MX); New Zealand (NZ); Philippines (PH); Russian Federation (RU); Sweden (SE); Slovenia (SI); Slovakia (SK); Suriname (SR); Thailand (TH); Taiwan (TW); United States (US); South Africa (ZA)

Method: LLM simulated data

- A set of 30 personas was generated using demographic distribution in the ISSP dataset. The following variables were included:
 - sex
 - age
 - education level
 - working status



- 30 peoples
- use the same set of persona across all simulations



Method: LLM simulated data

- 2 methods for generate LLM-simulated social network [1]
- global method
 - provide the LLM agent with a list of persona(E.g. 30 people) and ask it to generate a social network among those 30 person.
- local method:
 - step1. give the LLM one person at a time, along with information about other remaining 30-1 personas
 - step2. asking which of these 30-1 people the agent want to be friend with
 - repeat step 1 and 2 30 times
 - combine all response from 30 agents and create a social networks
- In each country, we apply both local and global methods to generate 30 networks with prompts written in their local language.
 - Each country has two sets of 30 networks, with each network containing 30 agents.

Local Method Prompt (benchmark)

- System prompt:
 - You are a typical adult.
 - Task ↓
 - • After silent reasoning, list EVERY person in the roster you would be close friends with. By "friends," we mean people you regularly interact with and trust enough to discuss personal matters or ask small favors.
 - • Output one line of IDs separated by comma + single space: ID1, ID2, ID3, ...
 - • Use ONLY IDs from the roster. Do NOT include your own ID. No extra text.
 - Think step-by-step internally; reveal ONLY the final ID list. **DO NOT ADD ANY ADDITIONAL WORDS**
- User prompt:
 - YOUR PROFILE
 - ID: 24
 - 24. female, age 69, Upper secondary, In paid work
 - ROSTER:
 - ID, Sex, Age, Education, Working Status
 - 2. male, age 48, Lower secondary, Retired
 - 28. female, age 23, Lower level tertiary, In paid work
 - 13. male, age 77, Lower level tertiary, Retired

Global Method Prompt (benchmark)

- System prompt:
 - You are a network generator.
 - Task ↓
 - • Using ONLY the people in the roster, create a realistic undirected friendship network.
 - By "friends," we mean people you regularly interact with and trust enough to discuss personal matters or ask small favors.
 - • Output each friendship as: ID1, ID2(comma + single space).
 - • One pair per line. No headers, no commentary, no duplicates, no self-ties.
 - Think silently; reveal ONLY the list of pairs.
- User prompt:
 - ROSTER:
 - ID, Sex, Age, Education, Working Status
 - 2. male, age 48, Lower secondary, Retired
 - 28. female, age 23, Lower level tertiary, In paid work
 - 13. male, age 77, Lower level tertiary, Retired

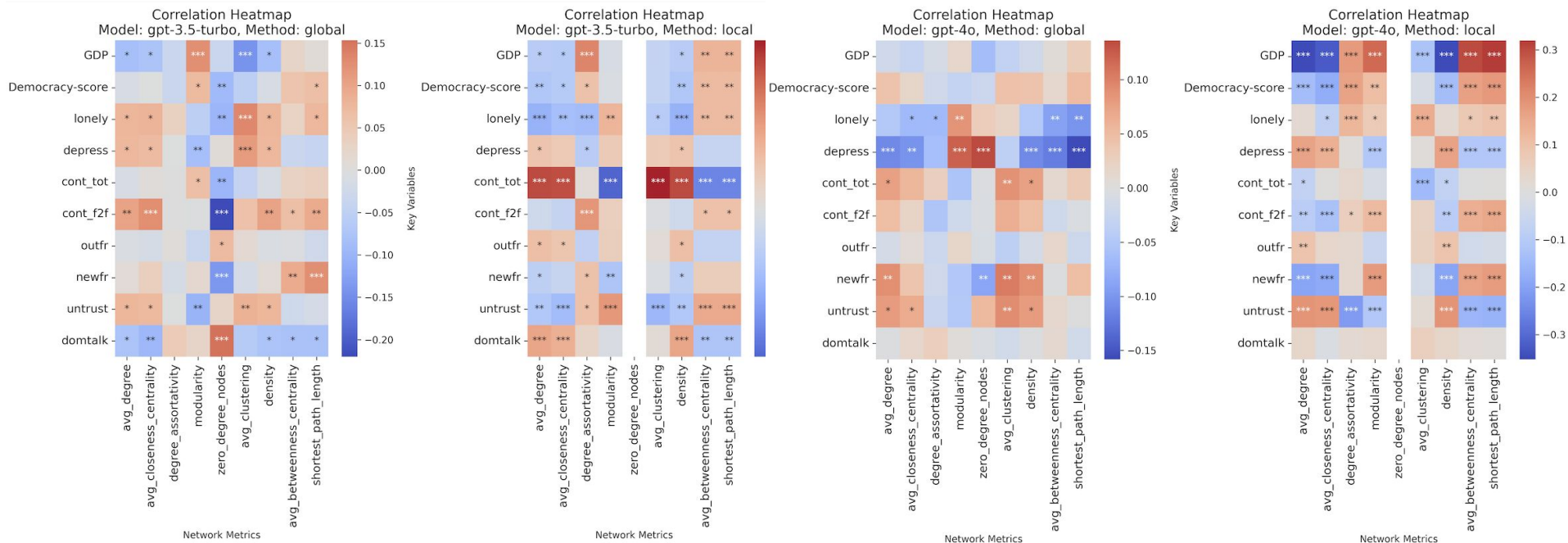
Local Method Prompt (Country specific, Spain)

- System prompt:
 - Eres un adulto típico que vive en España.
 - Tarea ↓
 - Después de pensar en silencio, enumera a todas las personas del listado con las que llegarías a ser amigo íntimo. Por «amigos» nos referimos a aquellas personas con las que interactúas con frecuencia y en las que confías lo suficiente como para hablar de asuntos personales o pedir pequeños favores.
 - • Salida: una línea con los IDs separados por una coma y un solo espacio: ID1, ID2, ID3, ...
 - • Usa únicamente los IDs del listado. No incluyas tu propio ID. Ningún otro texto adicional.
 - Piensa paso a paso en privado; muestra solo la lista final de IDs; no añadas ningún otro texto. Por favor, asegúrate de que el texto dentro de los paréntesis permanezca sin cambios.
- User prompt:
 - Tu perfil personal
 - ID: 12
 - 12. Hombre, 44 años, educación superior (avanzada), trabajo remunerado, vive en Spain
 - Lista:
 - ID, sexo, edad, nivel educativo, situación laboral, país
 - 16. Mujer, 34 años, educación secundaria, trabajo remunerado, Spain
 - 20. Mujer, 38 años, educación secundaria, trabajo remunerado, Spain
 - 18. Mujer, 37 años, educación postsecundaria (no universitaria), trabajo remunerado, Spain

Global Method Prompt (Country Specific, Spain)

- System prompt:
 - Eres un generador de redes.
 - Tarea ↓
 - Usando solo las personas del listado, crea una red realista y no dirigida de amistades para Spain.
 - El término “amigos” se refiere a aquellas personas con las que interactúas con frecuencia y en quienes confías lo suficiente como para hablar de asuntos personales o pedir pequeños favores.
 - • Salida: cada relación de amistad debe tener el formato:
 - ID1, ID2 (coma + un solo espacio).
 - Ejemplo de salida:
 - 6, 22
 - 7, 13
 - 8, 10
 - Una pareja por línea, sin título, sin comentarios, sin repeticiones ni autoenlaces.
 - Piensa en silencio; solo muestra la lista de pares correctos.
 - Asegúrate de que todo el texto dentro de las llaves se mantenga sin cambios.
- User prompt:
 - Lista:
 - ID, sexo, edad, nivel educativo, situación laboral, país
 - 16. Mujer, 34 años, educación secundaria, trabajo remunerado, España
 - 20. Mujer, 38 años, educación secundaria, trabajo remunerado, España
 - 18. Mujer, 37 años, educación postsecundaria (no universitaria), trabajo remunerado, España
 - 23. Mujer, 54 años, educación postsecundaria (no universitaria), trabajo remunerado, España

correlation between country level variable in real world data set (x-axis) and LLM simulated network in specific country context (y=axis)



- cont_tot=In contact with how many people on a typical weekday
- cont_f2f=Face-to-face contacts on a weekday

- outfr=How often go out with friend.
- newfr=How often make new friends when going out

- untrust=People can not be trusted or cannot be too careful in dealing with people
- domtalk=One person dominates conversation with friends

Reference

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Appendix: translation prompt

- Use chat gpt for translating prompt template file
 - Model: gpt-4o
- System Prompt:
 - You are a careful translation assistant who preserves code syntax.
- User prompt:
 - Translate the following into {target_lang}. Make sure your language style matches the main stream language style used in {country}. Please preserve all code-like expressions such as {{{{...}}}} and {{%...%}}:\n\n{text}. Make sure all text inside the brackets unchanged.

Method: Country level variables

- newfr: How often make new friends when going out
- trust: People can be trusted or cannot be too careful in dealing with people
- domtalk: One person dominates conversation with friends
- depress: Felt unable to overcome difficulties, unhappy and depressed (last 4 weeks)
- GDP:
 - GDP data from World Bank
- Democracy-score:
 - Democracy index from Our World